

Thoughtful use of AI: If you didn't know how to code, how would you use existing LLMs to accomplish this task? Write a brief (1 paragraph) description of how this could be done

If we weren't allowed to code and instead had to use existing LLMs to predict the next character, we could accomplish this by providing the context with LLM prompting. Specifically, we would prompt an LLM to give us exactly 3 characters and we could present those 3 characters to the astronaut. For added performance, we could prompt the model to do internal reasoning. One addition we could make was adding prompting involving languages of this astronaut (if this is information we have access to).

Benchmarking: Implement your ideas on using existing models and write a brief (1-2 paragraph) report on the results. You will need to use more than just the example data given in the github repo for full credit.

We benchmark Qwen/Qwen3-0.6B on a held out dataset of 1000 examples from 20 languages (see details below). Our model (using a character-gram with language detection) scored a 0.583 accuracy on this test set. The pretrained Qwen model scored a 0.307 accuracy on this test set.

One thing we observed was that the Qwen model had a very difficult time outputting exactly 3 characters out of the box, and even then still struggled to output the correct character, whereas our model had much better character control. Qwen also detects language far worse than our model (see examples below). We also theorize that should we force 3 characters to be outputted, the reasoning capabilities of these models will be constrained.

The languages are from `arabic (ar), bulgarian (bg), german (de), modern greek (el), english (en), spanish (es), french (fr), hindi (hi), italian (it), japanese (ja), dutch (nl), polish (pl), portuguese (pt), russian (ru), swahili (sw), thai (th), turkish (tr), urdu (ur), vietnamese (vi), and chinese (zh)`.

This is the test dataset of the hugging face dataset `papluca/language-identification`.

Sample failures from our model.

Incorrect prediction: 'etc' for correct char 'h'

Incorrect prediction: 'zkw' for correct char 'c'

Incorrect prediction: '्आम' for correct char 'न'

Incorrect prediction: 'त ं' for correct char 'स'

Sample Qwen failures:

Incorrect prediction: '内容对考试没帮助,和实际考试的热点相背离,' for correct char 'स'

Incorrect prediction: 'سکول آف میڈیسن مہم یا ڈاکٹر فیلڈز یا ا اگر آپ' for correct char 'o'

Incorrect prediction: '=== ' for correct char 'ل'